Engineering Ethics

When buildings fail



Learning objectives

01 Describe the key facts and timelines of the Grenfell Tower and Wang Fuk Court fires.

02 Explain how material choices and structural decisions contributed to the rapid spread of fire in both cases.

03 Compare the common factors across both disasters: flammable exterior materials, cost-cutting, ignored warnings, and regulatory failures.

04 Evaluate the ethical responsibilities of engineers, designers, manufacturers, and regulators when safety is compromised, and apply this understanding to your own design practice.

Grenfell Tower: the facts

14 June 2017

- 24-storey social housing tower in North Kensington, London; built in 1974.
- Refurbishment in 2012-16 added ACM cladding with a combustible polyethylene (PE) core.
- Fire reached the roof in about 30 minutes.
- 72 people were killed.

Task 1: Read and highlight materials, fire spread, and failures or cost-cutting.

Grenfell: how fire spread

Follow the material and design decisions

- ACM panels ignited; their polyethylene core was the principal fuel.
- The cavity and combustible foam insulation formed part of the external wall system. Rapid spread was not simply a cavity “chimney effect”.
- Fire entered flats through window assemblies, including combustible components.
- Compartmentation failed: fire spread between flats.

Wang Fuk Court: the facts

26 November 2025 · deaths confirmed 15 January 2026

- Eight 31-storey residential towers in Tai Po, Hong Kong; built in 1983.
- Facade renovation: bamboo scaffolding and non-compliant protective netting.
- Expanded polystyrene (EPS) foam boards covered windows.
- 168 people were killed, including one firefighter; seven towers were engulfed.

Task 1: Continue reading and highlighting the second case study.

Wang Fuk Court: fire spread

Reported mechanisms and concerns · see teacher guide, page 9

- Reports described netting igniting and spreading fire to EPS foam boards.
- Burning materials and damaged windows allowed fire into flats.
- Renovation materials created a continuous vertical fuel path, enabling rapid upward spread.
- Residents had reportedly warned about foam flammability months before.

Compare: place and buildings

Two cases, eight years apart

Factor	Grenfell Tower	Wang Fuk Court
Location	North Kensington, London, UK	Tai Po, Hong Kong, China
Building type	One 24-storey social housing tower; built 1974	Eight 31-storey residential towers; built 1983
Fatalities	72 people killed	168 people killed, including one firefighter

Compare: materials and spread

Both fires breached the exterior and entered homes

Factor	Grenfell Tower	Wang Fuk Court
Exterior fire fuel	ACM panels with a PE core; combustible PIR and phenolic foam insulation	Non-compliant scaffolding netting and EPS foam boards
Fire spread	Rapid external spread, driven mainly by the ACM core; fire entered through windows	Vertical spread through renovation materials; damaged windows let fire into flats

Compare: cost-cutting

What happens when cost decisions compromise safety?

Grenfell Tower

£293,368

A proposed saving for aluminium in place of zinc cladding with cassette fixing.

Not a price comparison between ACM fire ratings.

Wang Fuk Court

HK$105,800

2,300 rolls × (HK$100 − HK$54). About HK$106,000 using reported netting prices.

An illustrative price difference, not an audited total saving.

Compare: warnings and regulation

Distinguish Inquiry findings from reports and allegations

Factor	Grenfell Tower	Wang Fuk Court
Warnings	Residents repeatedly raised concerns; more than 40 safety issues flagged in 2016	Residents reportedly demonstrated flammability; 16 inspections did not avert the disaster
Regulation	Manufacturers falsified test data; regulation and building control failed	Reports and allegations of inspections missing non-compliant materials and falsified certificates

The pattern of failure

Five structural similarities between the two disasters

1	**Flammable exterior materials**	Combustible materials on the outside of both buildings fed the fires.
2	**Rapid upward spread**	Vertical fuel paths enabled rapid spread; the two external systems differed.
3	**Fire entry through windows**	Exterior fire breached flats through windows in both cases.
4	**Cost-cutting over safety**	Material choices were shaped by cost. Use the qualified figures on slide 9.
5	**Ignored warnings and failed regulation**	Residents raised concerns that were dismissed. Inspections and safety assurance failed; some Wang Fuk Court allegations remain under investigation.

Compare and connect

Which similarity is most significant, and why?

Task 2 · Comparison table

Work in pairs. Use the case studies and slides to begin the seven questions on page 3.

Task 3 · Venn diagram

On page 4, identify what is unique to each fire and what is shared by both. Aim for at least three points in each section.

Begin in the 17-23 minute comparison phase. Complete unfinished work after the lesson.

Engineering ethics

What responsibilities do engineers and designers have?

Duty of care	Professionals must prioritise the safety of building occupants above all other considerations.
Honesty and transparency	Test results, material properties, and safety data must never be falsified or concealed.
Accountability	Every person in the design and construction chain bears responsibility for the safety of the final product.
Professional competence	Engineers must ensure they fully understand the fire performance of the materials they specify.
Whistleblowing	Professionals have a moral obligation to raise concerns when they identify safety risks, even if it is commercially inconvenient.

Discuss: responsibility and choices

Use evidence from both case studies

- Should an engineer refuse to use a material they believe is unsafe, even if their employer insists?
- Who bears the most responsibility: the manufacturer who sold the product, the designer who specified it, or the regulator who approved it?
- If a cheaper material passes the minimum legal standard but a safer alternative exists, which should be used?

Discuss: changing practice

Connect engineering ethics to your own D&T work

- How can building regulations be improved to prevent similar disasters?
- What lessons should D&T students take from these cases into their own design practice?

Task 4: the key question

Evaluative writing · student activity sheet, page 5

What do the Grenfell Tower and Wang Fuk Court fires tell us about the ethical responsibilities of engineers, designers, manufacturers, and regulators when making decisions about building materials and safety?

Evaluate and write

33-44 minutes of the lesson

6-8 sentences · 11 minutes

Answer the key question using evidence from BOTH case studies. Work individually or in pairs.

Structure your response

Use the sentence starters and connectives on page 5. Consider different responsibilities and connect your conclusions to your own design practice.

What have we learned?

Key takeaways

- Engineers and designers have a duty of care that extends beyond meeting minimum legal requirements.
- Material selection has life-or-death consequences - understanding fire performance is essential.
- Cost-cutting is not just a financial decision; it is an ethical one.
- Systemic failures (not single mistakes) cause disasters - responsibility is shared across a chain.
- Regulations only protect people if they are properly enforced and honestly applied.

Exit ticket

Your judgement, supported by a reason

If you were an engineer asked to use a material you believed was unsafe, what would you do and why?

Write your answer on a sticky note or scrap paper. You have 2 minutes.

Acknowledgements and sources

Author
Wonyong Park (University of Southampton)

Reviewers
Susan Crawford and Laura Hanley

Project and artwork
The Grenfell Curriculum · grenfellcurriculum.com

Full reading list, dated factual updates and the retained 2017 ethical-principles reference: teacher guide, pages 7-9.

Edition updated 17 October 2026 · No speaker notes